

$\|f - f_M\|_2^2 \lesssim$	Sobolev 1-D	piecewise 1-D	Sobolev 2-D	BV 2-D	cartoon 2-D
Fourier, L	$M^{-2\alpha}$	M^{-1}	$M^{-\alpha}$	$M^{-1/2}$	$M^{-1/2}$
Fourier, NL	$M^{-2\alpha}$	M^{-1}	$M^{-\alpha}$	$M^{-1/2}$	$M^{-1/2}$
Wavelets, L	$M^{-2\alpha}$	M^{-1}	$M^{-\alpha}$	$M^{-1/2}$	$M^{-1/2}$
Wavelets, NL	$M^{-2\alpha}$	$M^{-2\alpha}$	$M^{-\alpha}$	M^{-1}	M^{-1}

Cartoon images: adaptive triangulations $O(M^{-2})$; curvelets $O(M^{-2}(\log M)^3)$.
L: linear; NL: nonlinear. Sobolev columns: H^α, $\alpha \geq 1$.
Use compatible wavelet bases and boundary conditions.
The BV class also has the uniform bound assumed in the chapter.